YAKEEN

NEET 2.0 2027

Physics | Lecture - 4

Vector

By- Manish Raj (MR Sir)



Todays Goal class

Revision of lecture-3 ✓

Question Practice

Todays Goal (H/W)

→ Sangh assig no (6) Basic math

→ " " no (1-32) Vectors

→ Sangha assig 1 to 6 (Basic math) ✓✓

1. Basic math → Summary lectr → Tomorrow morning → 1.5 hr

2. Sanghrsh Assignment no – 6 & 7 Todays will Uplad
Basic math

3. Sanghrsh)) no → 1 & 2 of vectr upladed.

Direction is more important than speed ⟶ MR. Sir.

Question (copy)

If angle between vector $\vec{A}$ and $\vec{B}$ is $30°$ then angle between

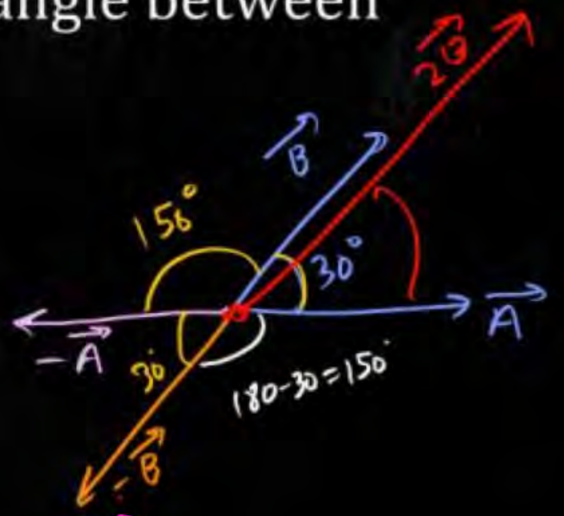
(a) $\vec{A}$ and $-\vec{B}$ $150°$ ✓

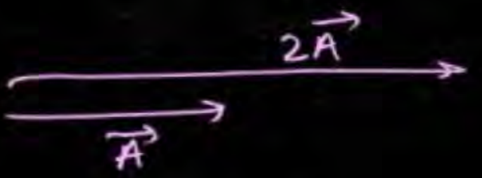
(b) $\vec{A}$ and $\vec{B}$ 30 ✓

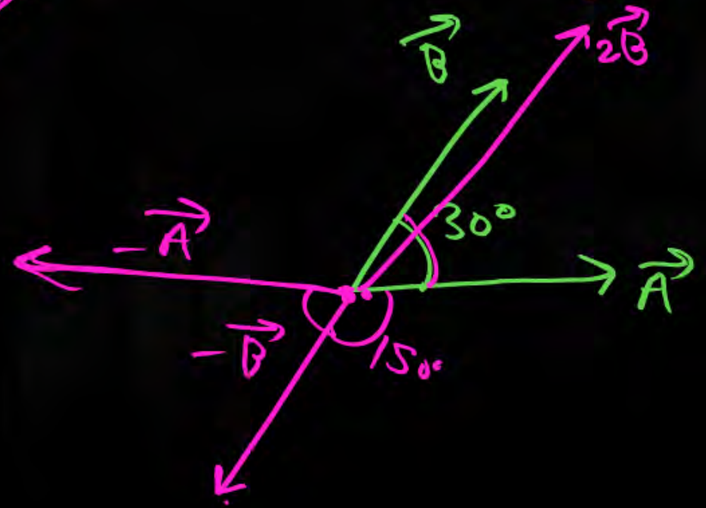
(c) $-\vec{A}$ and $-\vec{B}$ 30 ✓

(d) $\vec{A}$ and $2\vec{B}$ $30°$ ✓

(e) $2\vec{A}$ and $-3\vec{B}$ $150°$ ✓



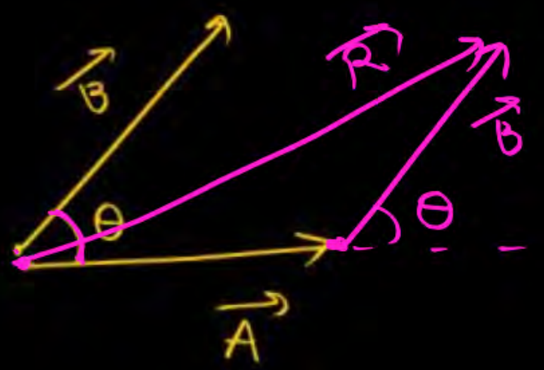


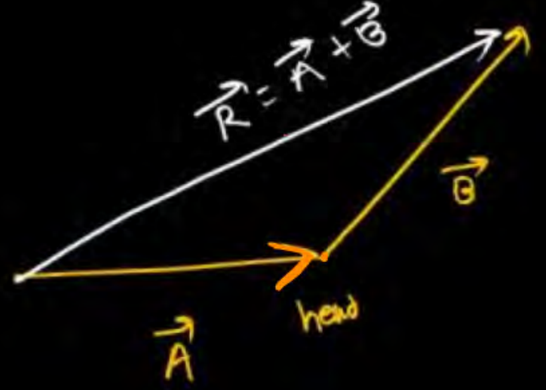


Poly gon - Law of vector addition

→ Koi bhi 1st vector lo, uske head se 2nd vector start karo, then 1st ke tail se 2nd ke head ko mila do; Wo vector dono ka resultant hoga.

→ graphical method of vector addition





$$\vec{R} = \vec{A} + \vec{B} = \vec{B} + \vec{A}$$



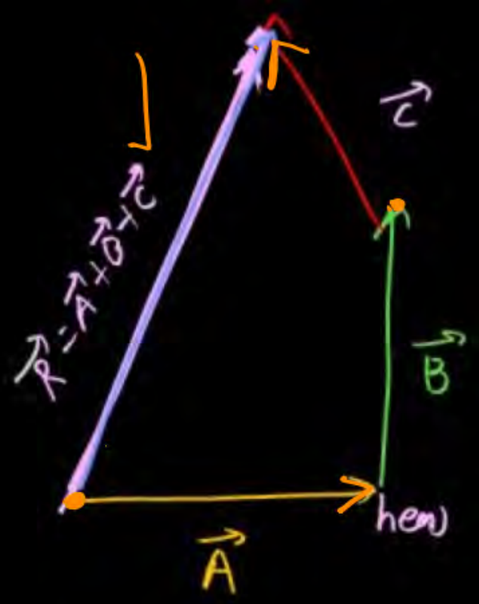
MR* BOX

Angle nikalne ke liye tail-tail or head-head ko ek sath rakho, vector add karne ke liye 1st ke head se 2nd ka tail start karo.

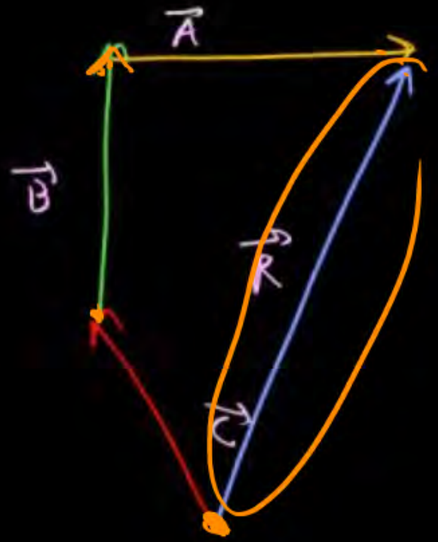
Ek bat or smajh lo jin 2-3 vector ko add kar rahe ho usme draw karne ka order kuch bhi le skte ho niche wale diagram se samjho. In all 3-diagram $\vec{R}$ is same.

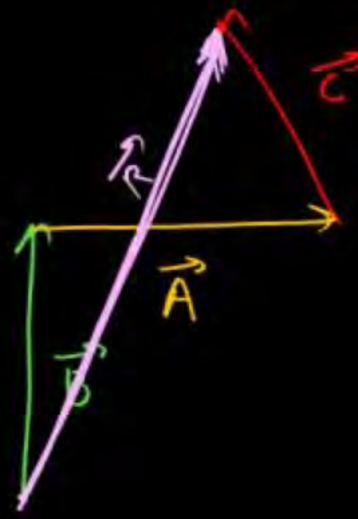
$$\vec{R} = \vec{A} + \vec{B} + \vec{C} = \vec{B} + \vec{C} + \vec{A} = \vec{C} + \vec{A} + \vec{B}$$

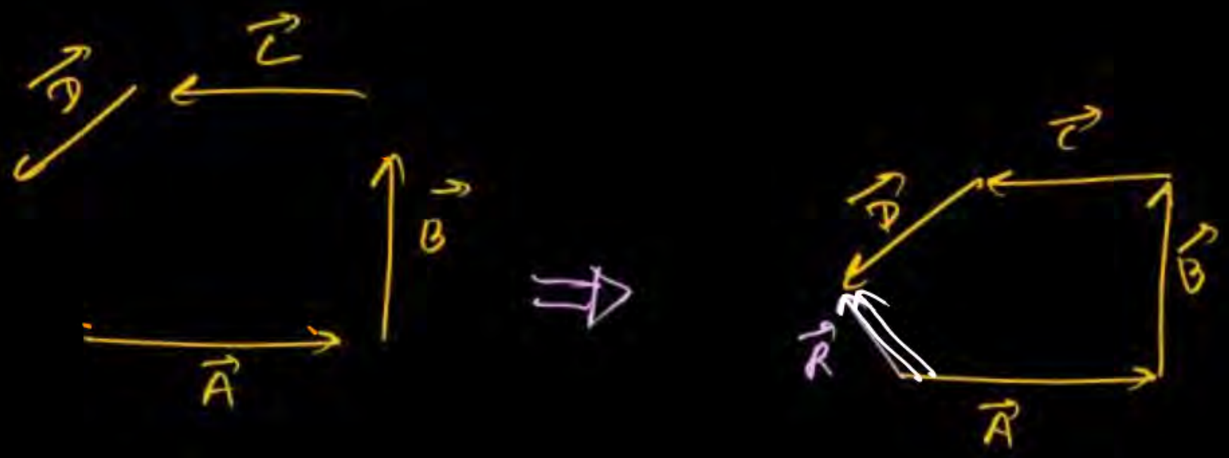


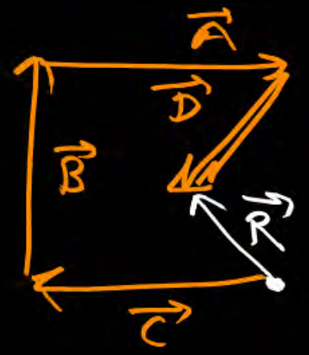


$$\vec{A}+\vec{B}+\vec{C}=\vec{B}+\vec{A}+\vec{C}=\vec{C}+\vec{B}+\vec{A}$$






D
C
B
A
D
C
B
R
A

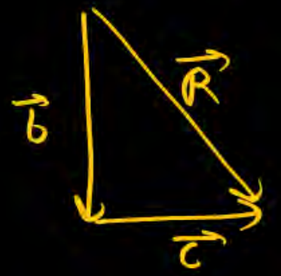

A
D
B
R
C

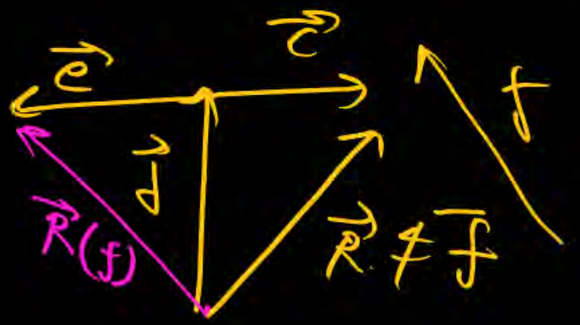
Question

No Copy!

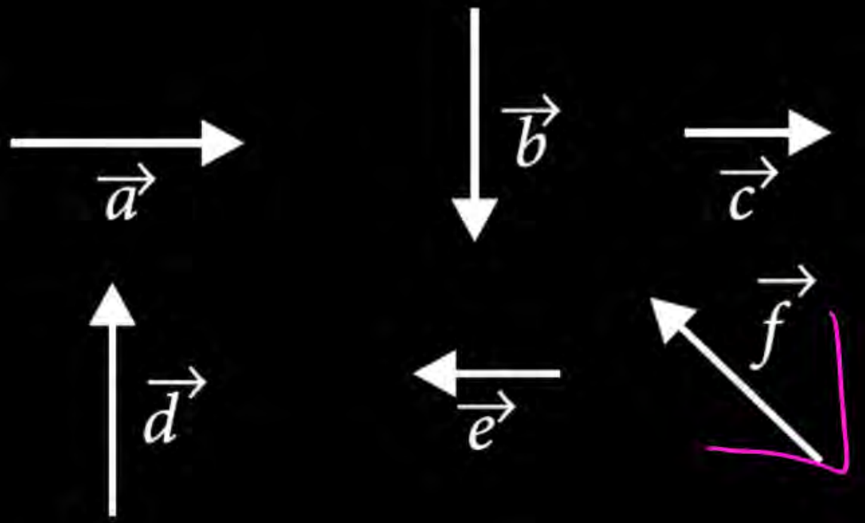
Six vectors, $\vec{a}$ through $\vec{f}$ have the magnitudes and directions indicates in the figure. Which of the following statements is true? **(2010)**

1. $\vec{b} + \vec{c} = \vec{f}$ ✗
2. $\vec{d} + \vec{c} = \vec{f}$ ✗
3. $\vec{d} + \vec{e} = \vec{f}$ ✓
4. $\vec{b} + \vec{e} = \vec{f}$ ✗

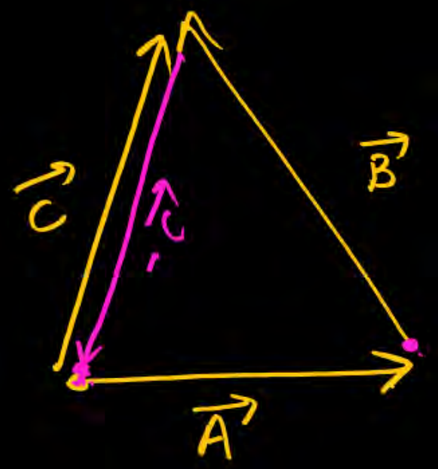








(Q) write relation b/w $\vec{A}$, $\vec{B}$ & $\vec{C}$ for given diagram



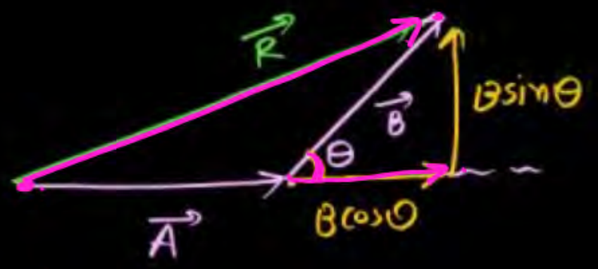
$$\vec{B} - \vec{C} + \vec{A} = 0$$ ✓✓

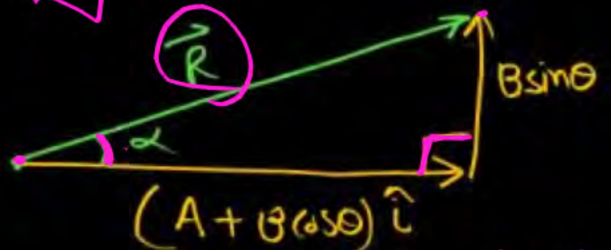
$$\vec{A} + \vec{B} = \vec{C}$$

$$\vec{A} + \vec{B} - \vec{C} = 0$$ ✓

Triangle Law of vector addition







$\vec{A}$ $\vec{B}$

$R = A + B$

$A = |\vec{A}|$ = Magnitude of $\vec{A}$

$B = |\vec{B}|$ = magnitude of $\vec{B}$

R = Magnitude of Resultant of $\vec{A}+\vec{B}$

θ : Angle b/w $\vec{A}$ & $\vec{B}$

$$R = \sqrt{(A+B\cos\theta)^2 + (B\sin\theta)^2}$$

$$= \sqrt{A^2 + B^2\cos^2\theta + 2AB\cos\theta + B^2\sin^2\theta}$$

$$R = \sqrt{A^2 + B^2(\cos^2\theta + \sin^2\theta) + 2AB\cos\theta}$$

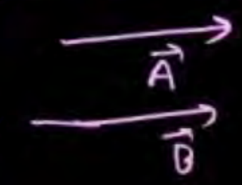
$$\tan\alpha = \frac{B\sin\theta}{A+B\cos\theta}$$

Magnitude of Resultant

MR Rattle

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

or $\theta = 0^\circ$

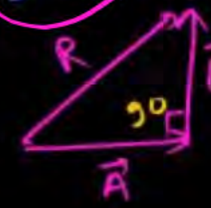


$$R = \sqrt{A^2+B^2+2AB\cos 0^\circ}$$

$$R = \sqrt{(A+B)^2}$$

$R = A + B$ Max

$\theta = 90^\circ$



$$R = \sqrt{A^2+B^2}$$

$\theta = 180^\circ$

$$R = \sqrt{A^2+B^2-2AB}$$

$$= \sqrt{(A-B)^2}$$

$R_{min} = A - B$

(Q)



$R = ??$

$$R = \sqrt{4^2 + 4^2 + 2 \times 4 \times 4 \times \cos 60^\circ}$$

$$R = \sqrt{16 + 16 + 2 \times 16 \times \frac{1}{2}}$$

$$= \sqrt{48} = \sqrt{16 \times 3} = \underline{\underline{4\sqrt{3}}}$$

Question

Two vector of magnitude 3 and 4 acting at different angle. Then find their resultant:

(1) $\theta = 0°$ ⟶ $R_{max} = 3+4 = 7$

(2) $\theta = 60°$ ⟶ $R = \sqrt{3^2+4^2+2\times3\times4\cos60°} = \sqrt{9+16+24\times\frac{1}{2}}$

$= \sqrt{37}$

(3) $\theta = 90°$ ⟶ $R = \sqrt{(3)^2+(4)^2+2\times3\times4\cos90°}$

$R = \sqrt{9+16} = 5$

(4) $\theta = 120°$ ⟶ $R = \sqrt{3^2+4^2+2\times3\times4\cos120°}$

(5) $\theta = 180°$

$R_m = \sqrt{3^2+4^2+2\times3\times4\cos(180)} = \sqrt{9+16-24}$

$= \sqrt{25-24} = \sqrt{1} = 1$

3 ← → 4

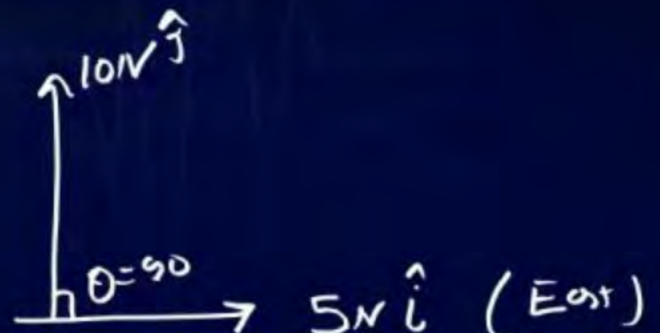
$4-3 = 1$

QUESTION

Two force $\vec{F}_1 = 5N$ due east and $F_2 = 10$ N due north then resultant of these two force is

1. $5\sqrt{5}$ N
2. 15 N
3. 5 N
4. $\sqrt{5}$ N





$R = \sqrt{(5)^2 + (10)^2}$

$= \sqrt{125}$

$= \sqrt{25 \times 5} = 5\sqrt{5}$

QUESTION

Find net force $\vec{F}_1 + \vec{F}_2$.

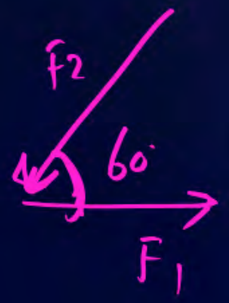
1. 5N
2. $5\sqrt{3}$N
3. $5\sqrt{2}$N
4. 0

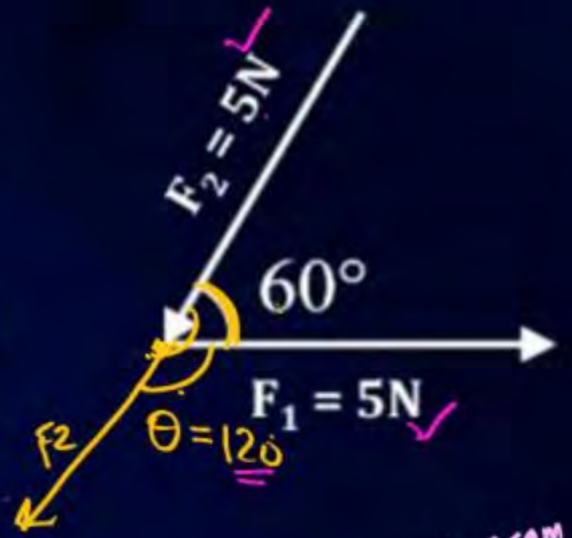
$R = \sqrt{A^2+B^2+2AB\cos\theta}$

$\vec{A}$ & $\vec{B}$ vector

$\vec{A} + \vec{B} = \vec{R}$ (diff ke sath)

A + B (sum of their magnitude)





$R = \sqrt{5^2+5^2+2\times5\times5\cos120^\circ}$

$= \sqrt{25+25+50\times(-\frac{1}{2})}$

$= \sqrt{50-25} = \sqrt{25} = 5$

$R = \sqrt{5^2+5^2+2\times5\times5\cos60^\circ}$

$= \sqrt{50+2\times25\times\frac{1}{2}} = \sqrt{75}$

NR scam

Ans: (1)

Question

Which pair of two vector can not give resultant of magnitude 4.

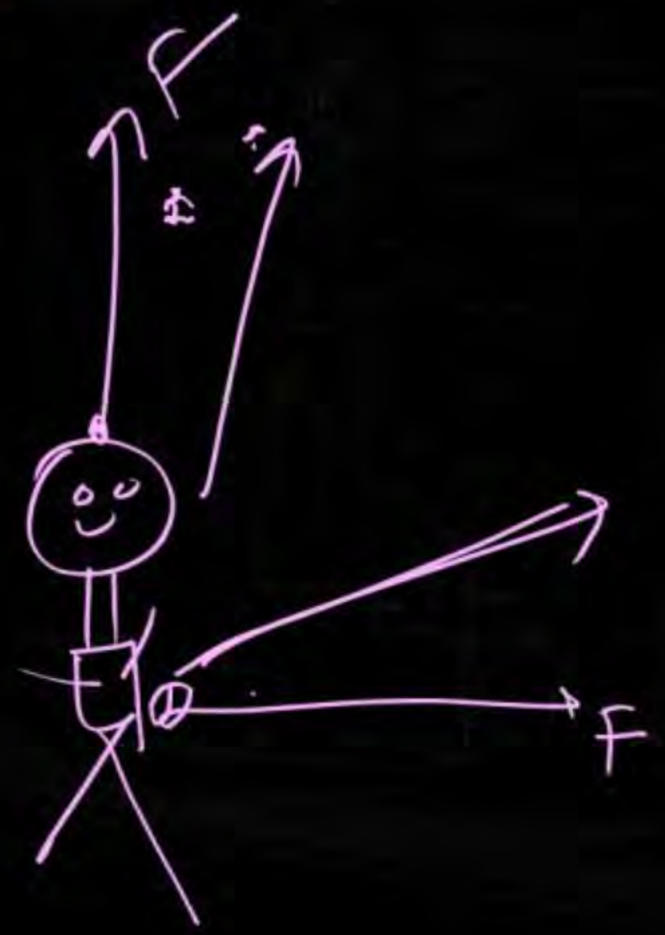
(a) 2, 4 → 4-2=2, 6 (min, max)

(b) 4, 4 → 4-4=0, 4+4=8

(c) 4, 9 → 9-4=5, 6, 7, 8, 9, 10, 11, 12, 13

(d) 1, 5 → 5-1=4, 5+1=6

14 ✗ 15 ✓



4, 2 } 6

Ans (c)

4, 2

$R_{max\ obt} = |\vec{A}| + |\vec{B}| = A + B$

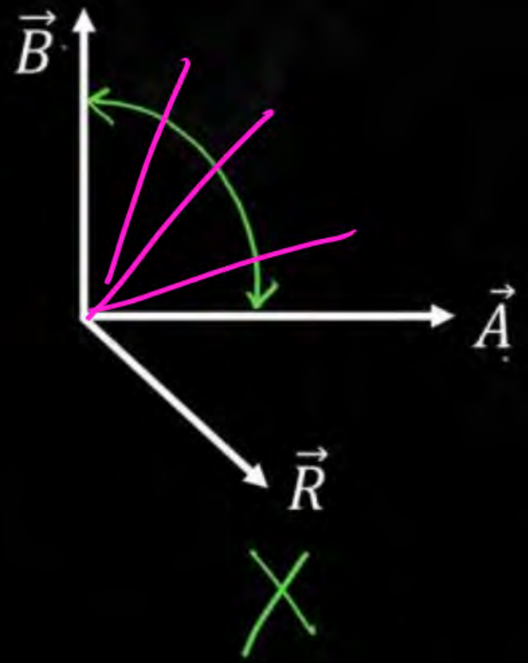
$R_{min\ obt} = |\vec{A}| - |\vec{B}|$ ✓

$|\vec{A}| - |\vec{B}| \leq R \leq |\vec{A}| + |\vec{B}|$

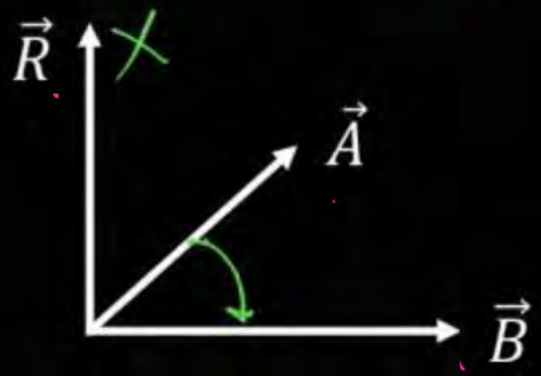
4, 2, 4-2=2

Question

Which of the following diagram is correct: for Reln $\vec{A}+\vec{B}=\vec{R}$



X



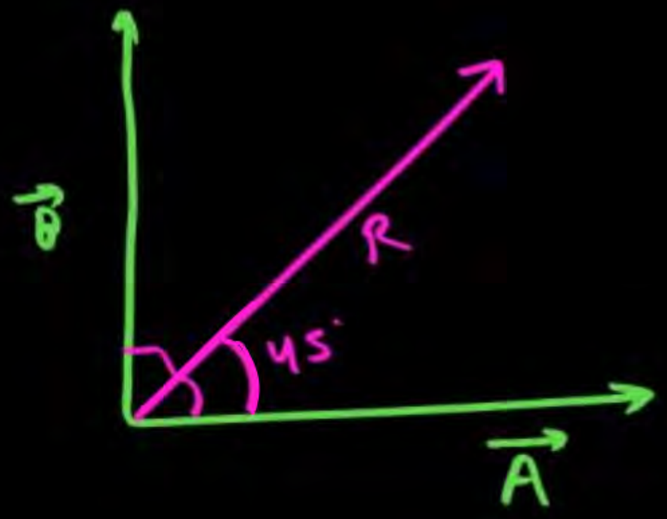
X

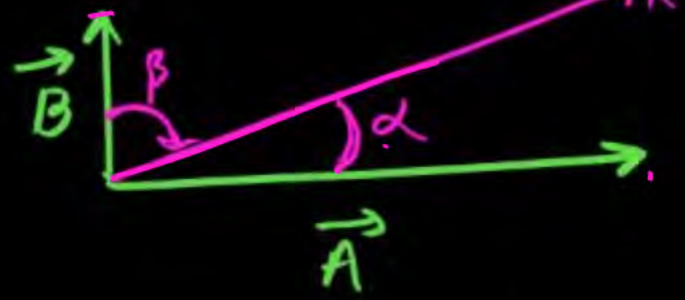


Correct

MR* BOX

2-vector ka resultant (Sum) unke bich me rahega, bade wale vector ke pass niche wale diagram se feel karo





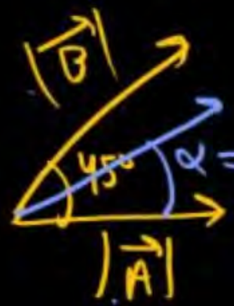
$\alpha < \beta$



$\beta < \alpha$

If $|\vec{A}| = |\vec{B}|$

AIEEE



If $|\vec{A}| = |\vec{B}|$ then find Angle b/w $\vec{R}$ & $\vec{A}$.

✓ Ans

Question

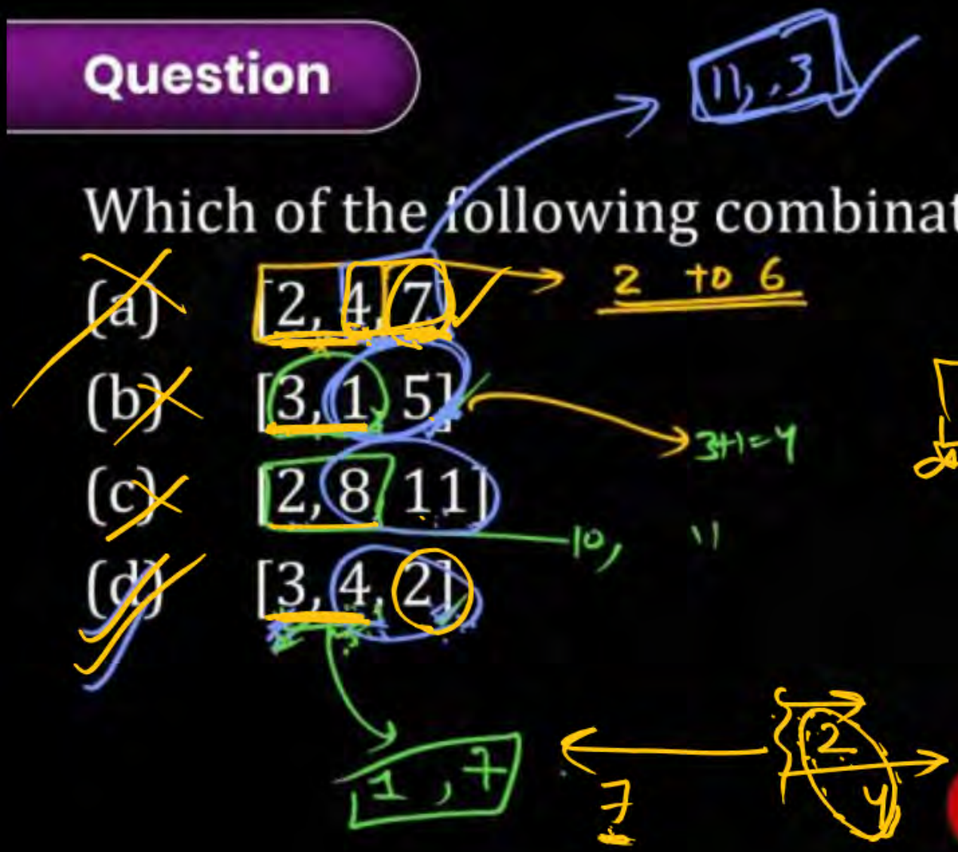
Which of the following combination of three force can give zero resultant?

(a) [2, 4, 7]

(b) [3, 1, 5]

(c) [2, 8, 11]

(d) [3, 4, 2]



Ans (d)

MR* BOX

Koi bhi 2-vector lo uske resultant ke range (A – B to A + B) me 3rd vector aaya to sum of all three vector zero ho sakta hai

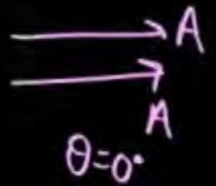
Copy # 9f Two vector of same magnitude $|\vec{A}| = |B|$ acting at angle θ then find Their Resultant

$$R = \sqrt{A^2 + A^2 + 2AA\cos\theta}$$

$$R = 2A\cos(\theta/2)$$ MR Ratta

$\theta = 0°$

$R = 2A$

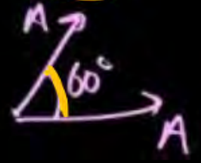


$\theta = 60°$

$R = 2A\cos(30)$

$= 2A\frac{\sqrt{3}}{2}$

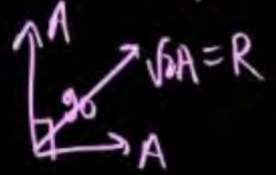
$= \sqrt{3}A$



$\theta = 90°$

$R = 2A\cos(45)$

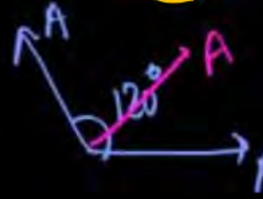
$= 2A \times \frac{1}{\sqrt{2}} = \sqrt{2}A$



$\theta = 120°$

$R = 2A\frac{1}{2}$

$R = A$



$\theta = 180°$ ✓

$R = 0$



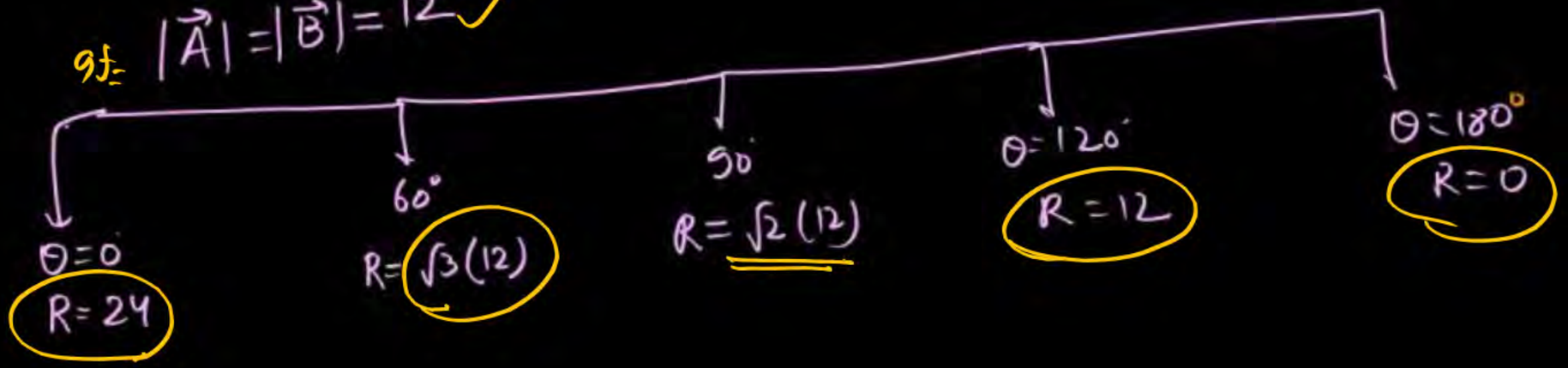
$R = 0$

$R = 0$ ✓



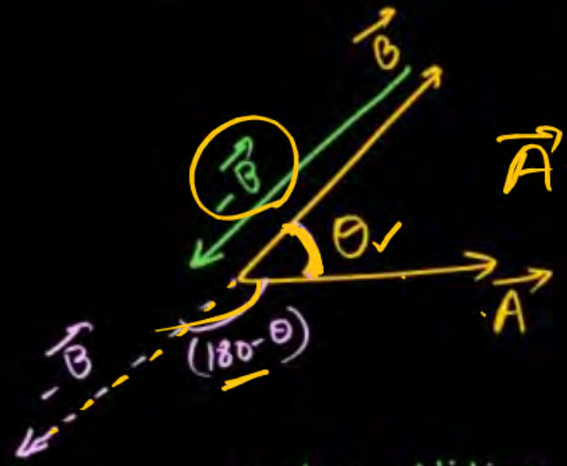
#

9) $|\vec{A}| = |\vec{B}| = 12$ ✓

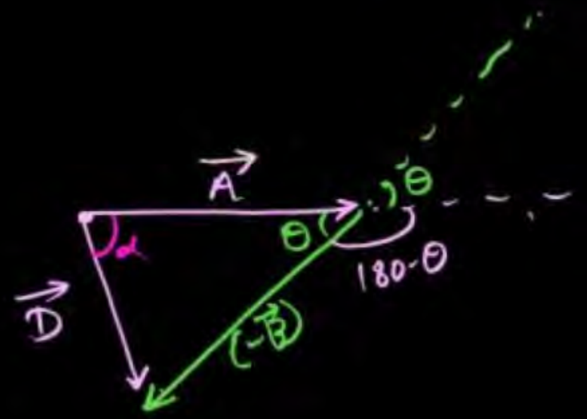


Vector subtraction

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$



$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$



$$D = \sqrt{A^2 + B^2 + 2AB\cos(180-\theta)}$$

$$D = \sqrt{A^2 + B^2 - 2AB\cos\theta}$$

$$D = \sqrt{A^2 + B^2 - 2AB\cos\theta}$$

$\theta = 90°$

$\theta = 0$

$$D = \sqrt{A^2 + B^2 - 2AB \times 1}$$

$$= \sqrt{(A-B)^2}$$

$$D = A - B \quad (\text{min})$$

$$D = \sqrt{A^2 + B^2}$$

$\theta = 180$

$$D = A + B$$

180°

Vector addition



$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

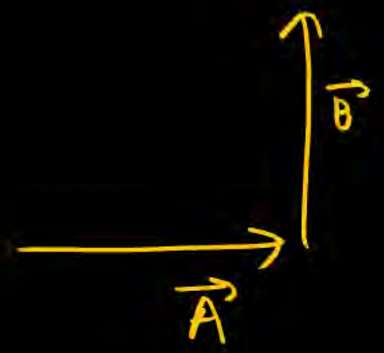
$$R = 2A\cos(\theta/2)$$

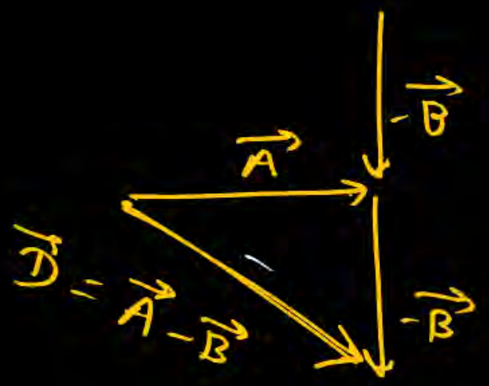
$$\cos(180-\theta) = -\cos\theta$$



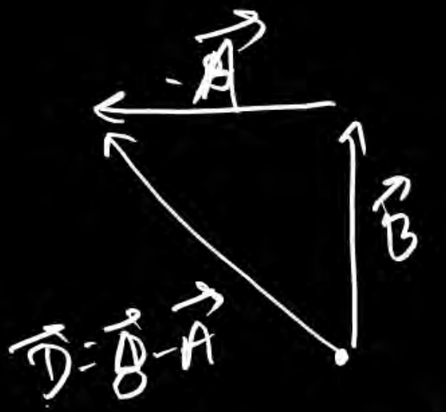
$$\tan\alpha = \frac{B\sin\theta}{A - B\cos\theta}$$

$\vec{A} - \vec{B} = ??$



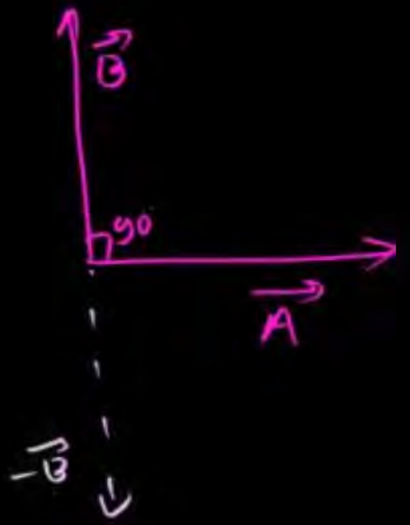


$\vec{B} - \vec{A} = \vec{B} + (-\vec{A})$



$\vec{A} + \vec{B} = \vec{B} + \vec{A}$ ✓

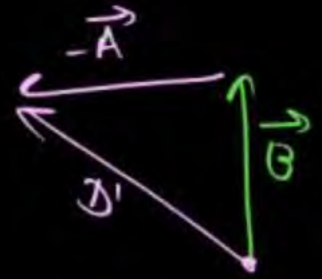
$\vec{A} - \vec{B} \neq \vec{B} - \vec{A}$ ✓



$\vec{R} = \vec{A} + \vec{B}$

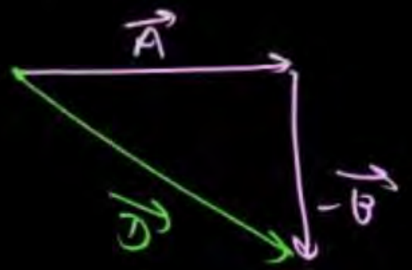


$\vec{D}' = \vec{B} - \vec{A} = \vec{B} + (-\vec{A})$



- $\vec{A} + \vec{B} = \vec{B} + \vec{A}$
- $\vec{A} - \vec{B} \neq \vec{B} - \vec{A}$

$\vec{D} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B})$



$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$A = B \checkmark$$

$$R = \sqrt{A^2 + A^2 + 2A \cdot A\cos\theta}$$

$$2A\cos\left(\frac{\theta}{2}\right)$$

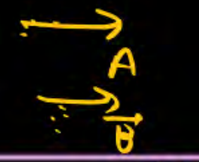
Two vector of same magnitude $|\vec{A}| = |\vec{B}| = A$ acting at angle θ then magnitude of their Resultant & magnitude of their Subtraction.

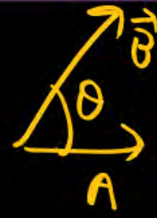
$R = 2A\cos\frac{\theta}{2}$ ✓

$D = 2A\sin(\theta/2)$ ✓

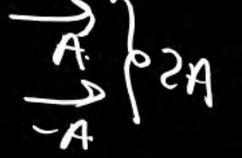
$\theta = 0$	$\theta = 60$	$\theta = 90°$	$\theta = 120$	$\theta = 180$
$R = 2A$ ✓	$R = \sqrt{3}A$ ✓	$R = \sqrt{2}A$ ✓	$R = A$ ✓	$R = 0$ ✓
$D = 0$ ✓	$D = A$ ✓	$D = \sqrt{2}A$ ✓	$D = \sqrt{3}A$ ✓	$D = 2A$ ✓

mp4









Vector Addition

2016, 2006, 2001

If $\vec{A} + \vec{B} = \vec{R}$ and $\vec{D} = \vec{A} - \vec{B}$ then angle between $\vec{A}$ + and $\vec{B}$ for given condition

$|\vec{A}| = |\vec{B}| = |\vec{R}| \rightarrow 120°$

$|\vec{A}| = |\vec{B}| = |\vec{D}| \rightarrow 60°$

$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}| \rightarrow 90°$

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta} \rightarrow R = 2A\cos(\theta/2)$

$D = \sqrt{A^2 + B^2 - 2AB\cos\theta} \rightarrow D = 2A\sin(\theta/2)$

If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is: (2016-I)

$|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$

1. 1°
2. 90°
3. 45°
4. 180°

Question

If $|\vec{A}+\vec{B}| = |\vec{A}| + |\vec{B}|$ then angle between A and B will be: (2001)

$R = A + B$

1. $90°$
2. $120°$
3. $0°$
4. $60°$

Question

For Two vector $\vec{A}$ and $\vec{B}$, vector addition $\vec{R} = \vec{A} + \vec{B}$ and vector Subtraction $\vec{D} = \vec{A} - \vec{B}$ then Angle between $\vec{A}$ and $\vec{B}$ for given condition.

(1) $|\vec{A} + \vec{B}| = |\vec{R}| = |\vec{A}| = |\vec{B}|$ → 120°

(2) $|\vec{A}| = |\vec{B}| = |\vec{A} - \vec{B}|$ → 60°

(3) $A^2 + B^2 = |\vec{A} - \vec{B}|^2 = D^2$ → $\sqrt{A^2+B^2} = D$ (90°)

(4) $A^2 + B^2 = R^2$ → $\theta = 90$

(5) $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$ → 90° ✓

(B) $|\vec{A} + \vec{B}| = R = |\vec{A}| + |\vec{B}|$ $R = A + B \Rightarrow 0°$

(7) $D = |\vec{A} - \vec{B}| = |\vec{A}| + |\vec{B}|$ → $D = A + B$ → 180°

(8) $|\vec{A} + \vec{B}| = R = |\vec{A}| - |\vec{B}|$ → $R = A - B$ → 180°

(9) $|\vec{A} - \vec{B}| = D = |\vec{A}| - |\vec{B}|$ → $\theta = 0°$

(10) $R = |\vec{A} + \vec{B}| = \sqrt{3}A = \sqrt{3}B$ → $\theta = 60°$

(11) $|\vec{A} - \vec{B}| = \sqrt{3}A = \sqrt{3}B$ → (120°)

Question

$|\vec{A}| = 1 \quad |\vec{B}| = 1$

If the sum of two unit vectors is a unit vector, then the magnitude of their difference is:

$\{|\vec{A}| = 1, \ |\vec{B}| = 1\} \longrightarrow |\vec{A}+\vec{B}| = |\vec{R}| = 1$

$\theta = 120^\circ$

$|D| = |\vec{A} - \vec{B}| = \sqrt{3} \times 1$

$D = \sqrt{1^2 + 1^2 - 2 \times 1 \times 1 \cos 120^\circ}$

$\sqrt{3}$

$D = 2A \sin(\theta/2)$

$= 2 \times 1 \sin(120^\circ/2)$

$= 2 \times 1 \frac{\sqrt{3}}{2} = \sqrt{3}$

Question

finding of unit vector :—

Find unit vector $\hat{A} = \frac{\vec{A}}{A}$

(1) $\vec{A} = \hat{i}$ ⟶ $\hat{A} = \hat{i} = \frac{i}{1} = \hat{i}$

(2) $\vec{B} = \hat{i} + \hat{j}$ $|\vec{B}| = \sqrt{1^2+1^2} = \sqrt{2}$ $\hat{B} = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$ ✓

(3) $\vec{C} = \hat{i} + \hat{j} - \hat{k}$ $\hat{C} = \frac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{3}}$ ✓

(4) $\vec{D} = \sin\theta\,\hat{i} - \cos\hat{j}$ $\vec{D} = \frac{\sin\theta\,\hat{i} - \cos\theta\,\hat{j}}{2}$ ✓

(5) $\vec{E} = 3\hat{i} + 4\hat{j}$ $\vec{E} = \frac{3\hat{i}+4\hat{j}}{5}$ ✓

$|E| = 5$

$\vec{A} = A\hat{A}$ unit vector

$|\vec{T}| = \sqrt{\sin^2\theta + (-\cos\theta)^2}$

$= \sqrt{\sin^2\theta + \cos^2\theta} = 1$

∴



$\vec{A} = A\hat{A}$

$\hat{A} = \frac{\vec{A}}{A}$ ✓

$\vec{C} = i + j - \hat{k}$

$|\vec{C}| = \sqrt{1^2 + 1^2 + (-1)^2}$

$= \sqrt{1+1+1}$

$= \sqrt{3}$

Copy

- Vector addition / subtraction in terms of component:

$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$

$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$

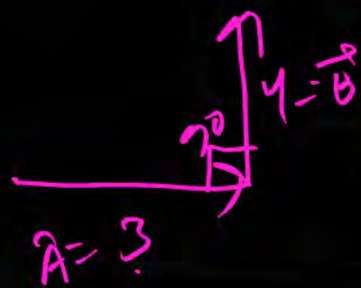
$\vec{A} - \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) - (B_x\hat{i} + B_y\hat{j} + B_z\hat{k}) = (A_x - B_x)\hat{i} + (A_y - B_y)\hat{j} + (A_z - B_z)\hat{k}$

$\vec{B} - \vec{A} = (B_x - A_x)\hat{i} + (B_y - A_y)\hat{j} + (B_z - A_z)\hat{k}$

$|\vec{A}| + |\vec{B}|$

↑ Sum of magnitude

$\vec{A} + \vec{B} = |\vec{A}| + |\vec{B}|$ → 0° Case



$\vec{B} + \vec{A} = 5$ ✓

7

$\vec{A} = 3$ $\vec{B} = 4$ → 7 ✓

Question

Copy

$\vec{A} = 4\hat{i} + 3\hat{j}$; $\vec{B} = 3\hat{i} - 4\hat{j}$ then find

(1) $\vec{A} + \vec{B} = (4i+3\hat{j}) + (3i-4\hat{j}) = 7i - \hat{j}$

(2) $\vec{A} - \vec{B} = \hat{i} + 7\hat{j}$

(3) $\vec{B} - \vec{A} = 3i-4\hat{j} - 4i-3\hat{j} = -i-7\hat{j}$

(4) $|\vec{A}| + |\vec{B}| = 5 + 5 = 10$

Question

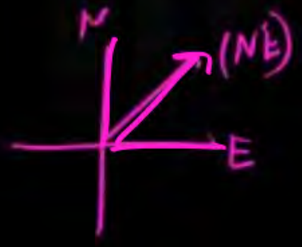
Initial velocity V = $10\hat{i}$ ↑ inecst after some time its velocity becomes $10\hat{j}$ m/s north then find change in velocity is?

$\vec{V_i} = 10\hat{i}$

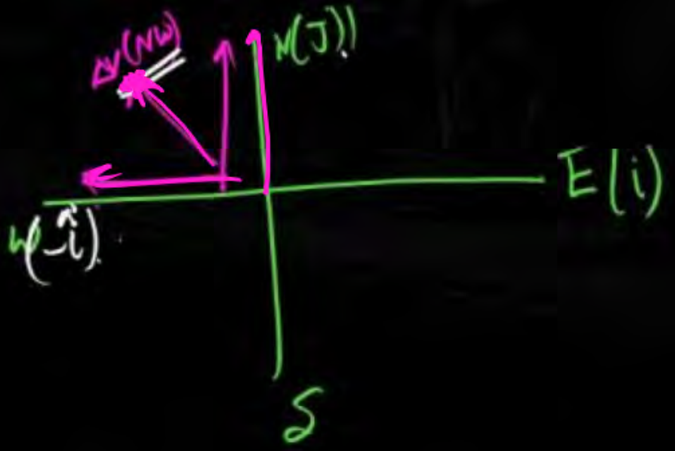
$\vec{V_f} = 10\hat{j}$

(1) # Sum of velocit $= \vec{V_i} + \vec{V_f}$

$= 10\hat{i} + 10\hat{j}$



Chage in velocity $\vec{V_f} - \vec{V_i} = 10\hat{j} - 10\hat{i}$



Question

Two vector $\vec{A} = 6\hat{i}$ and $\vec{B} = 8\hat{j}$ then

(1) $\vec{A} + \vec{B} =$ $6i + 8j$

(2) $\vec{A} - \vec{B} =$ $6i - 8j$

(3) $|\vec{A}| + |\vec{B}| =$ $6 + 8 = 14$

(4) $|\vec{A} + \vec{B}| =$??

$$|\vec{A} + \vec{B}| = |6i + 8j| = \sqrt{6^2 + 8^2}$$

$$= \sqrt{36 + 64} = 10$$

Question

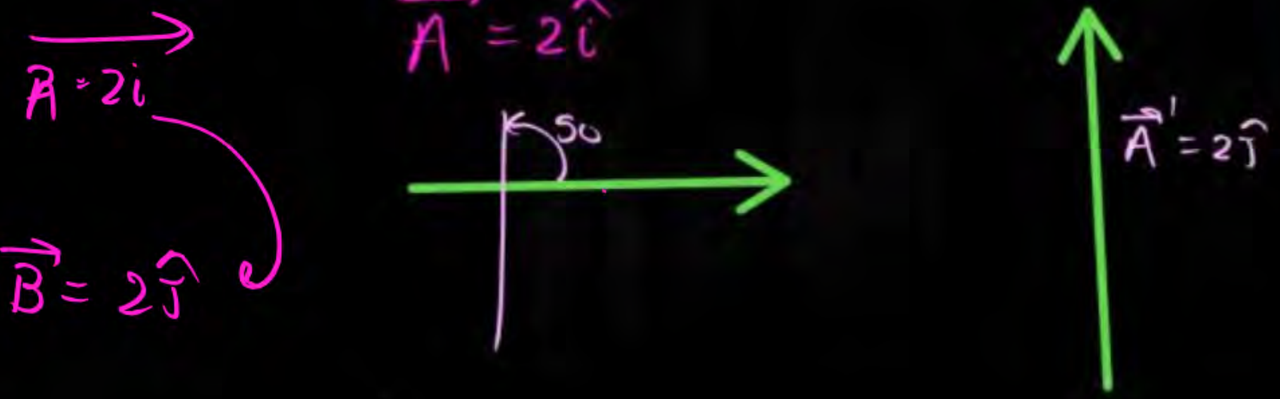
A vector $2\hat{i}$ rotated, then new vector will be:

(i) By $\theta = 90°$ (A.C) $= A' = 2\hat{j}$

(ii) By $\theta = 180°$ (A.C) $= A' = -2\hat{i}$

(iii) By $\theta = 270°$ (A.C) $= A' = -2\hat{j}$

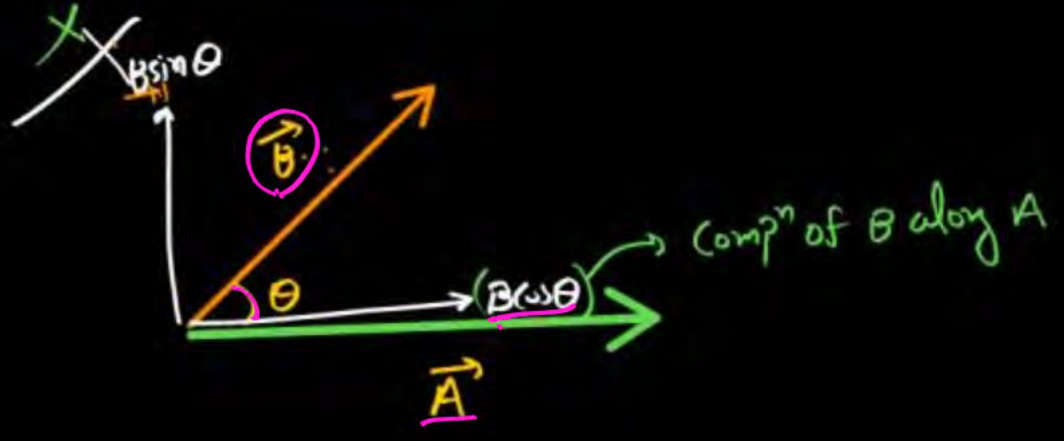
(iv) By $\theta = 360°$ (A.C) $= A' = 2\hat{i}$



MR* BOX

In rotation of vector magnitude remains same and direction changes.

Dot Product [Scalar Product]



$$\vec{A} \cdot \vec{B} = A B \cos\theta$$

$\vec{A}$ vecto

Angle b/w $\vec{A}$ & $\vec{B}$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Application of dot Product:-

(i) To check ⊥ vector

If $\vec{A} \cdot \vec{B} = 0$ hence $\vec{A} \perp \vec{B}$

(ii) To find Angle b/w vectors

$$\frac{\vec{A} \cdot \vec{B}}{A B} = \cos\theta$$

(iii) To find component of one vector along other.

$$\vec{A} \cdot \vec{B} = A (B\cos\theta)$$

$$B\cos\theta = \frac{\vec{A} \cdot \vec{B}}{A} = \text{Comp}^{n} \text{ of } B \text{ along } \vec{A}$$

(iv) To check unit vector

$\vec{A} \cdot \vec{A} = 1$

$\hat{i} \cdot \hat{i} = 1 \times 1 \cos 0 = 1$

$\hat{j} \cdot \hat{j} = 1$

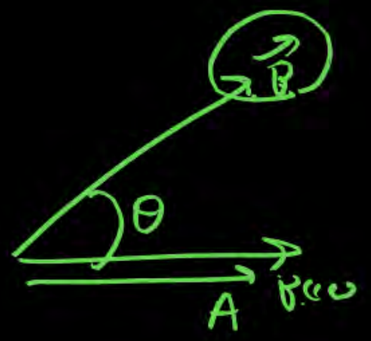
$\hat{k} \cdot \hat{k} = 1$

$\hat{i} \cdot \hat{j} = 0$

$\hat{i} \cdot \hat{k} = 0$

$\hat{k} \cdot \hat{j} = 0$

90°



$$\vec{A} \cdot \vec{B} = AB\cos\theta$$

$$B\cos\theta = \frac{\vec{A} \cdot \vec{B}}{A}$$ ✓

Comp^{n} of $\vec{A}$ along $\vec{B}$ = $\frac{\vec{A} \cdot \vec{B}}{B}$ ✓

Question

If $\vec{A} = 2\hat{i} + 3\hat{j} + \alpha\hat{k}$ and $\vec{B} = \hat{i} - 2\hat{j} + 4\hat{k}$ find α. If $\vec{A}$ is perpendicular to $\vec{B}$. (NEET)

$$2 - 6 + 4\alpha = 0$$

$$-4 + 4\alpha = 0$$

$$4\alpha = 4$$

$$\alpha = 1$$

QUESTION

If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $3\hat{i} - 4\hat{j} + \alpha\hat{k}$, then the value of α is [2005]

1. 1/2
2. -1/2
3. 3/4
4. -1

$\vec{A} \cdot \vec{B}$

$6 - 12 + 8\alpha = 0$

$-6 + 8\alpha = 0$

$8\alpha = 6$

$\alpha = \frac{6}{8} = \frac{3}{4}$

QUESTION

If vector $\vec{A} = \cos \omega t\, \hat{i} + \sin \omega t\, \hat{j}$ and $\vec{B} = \cos \frac{\omega t}{2}\, \hat{i} + \sin \frac{\omega t}{2}\, \hat{j}$ are function of time, then the value of t is at which they are orthogonal to each other is [2005]

Perpendicular

1. $t = \pi/\omega$
2. $t = 0$
3. $t = \pi/4\omega$
4. $t = \pi/2\omega$

$$\cos(\omega t) \cdot \cos\left(\frac{\omega t}{2}\right) + \sin(\omega t) \cdot \sin\left(\frac{\omega t}{2}\right) = 0$$

$$\cos(A - B) = \cos\left(\omega t - \frac{\omega t}{2}\right) = 0$$

$$\cos\left(\frac{\omega t}{2}\right) = 0$$

$$\frac{\omega t}{2} = \frac{\pi}{2}$$

$$t = \pi/\omega$$

Ans: (1)

QUESTION

The angle between the two vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ will be

[1994]

1. 90°
2. 180°
3. zero
4. 45°

$\vec{A} - \vec{B}$: تعامد

AB

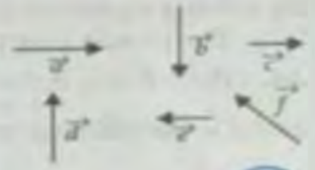
Question

If $\vec{A} = 2\hat{\imath} + 6\hat{\jmath} + 3\hat{k}$ and $\vec{B} = 4\hat{\imath}$. Find angle between $\vec{A}$ and $\vec{B}$.

$$\vec{A} \cdot \vec{B} = A \cdot B \cos\theta$$

3.4 Addition and Subtraction of Vectors-Graphical Method

2. Six vectors, $\vec{a}$ through $\vec{f}$ have the magnitudes and directions indicated in the figure. Which of the following statements is true?



(a) $\vec{b}+\vec{c}=\vec{f}$ (b) $\vec{d}+\vec{c}=\vec{f}$

(c) $\vec{d}+\vec{e}=\vec{f}$ (d) $\vec{b}+\vec{e}=\vec{f}$ *(2010)*

3.5 Resolution of Vectors

3. If a unit vector is represented by $0.5\hat{i}-0.8\hat{j}+c\hat{k}$ then the value of c is

(a) $\sqrt{0.01}$ (b) $\sqrt{0.11}$

(c) 1 (d) $\sqrt{0.39}$ *(1999)*

3.6 Vector Addition-Analytical Method

4. If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is

(a) 45° (b) 180°

(c) 0° (d) 90° *(NEET-I 2016)*

5. The vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$. The angle between the two vectors is

(a) 45° (b) 90°

(c) 60° (d) 75° *(2006, 1996, 1991)*

6. If $|\vec{A}+\vec{B}| = |\vec{A}|+|\vec{B}|$ then angle between A and B will be

(a) 90° (b) 120° (c) 0° (d) 60°. *(2001)*

7. The magnitude of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 3, 4 and 5 units respectively. If $\vec{A}+\vec{B}=\vec{C}$, the angle between $\vec{A}$ and $\vec{B}$ is

(a) $\pi/2$ (b) $\cos^{-1}(0.6)$

(c) $\tan^{-1}(7/5)$ (d) $\pi/4$. *(1988)*

$\sqrt{(0.5)^2+(0.8)^2+c^2}=1$

$0.25+0.64+c^2=1$

$\vec{A}=0.5i-0.8j+ck$

$\vec{A}=0.5i-0.8j+ck$

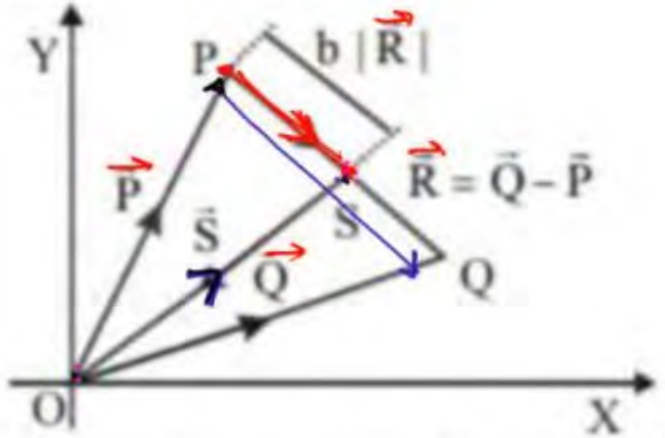
$\vec{A}\cdot\vec{A}=1$





$5=\sqrt{3^2+4^2+2\times3\times4\cos\theta}$

5. Three vectors $\vec{P}, \vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $b|\vec{R}|$. The general relation among vectors $\vec{P}, \vec{Q}$ and $\vec{S}$ is **[Adv. 2017]**



(a) $\vec{S} = (1-b)\vec{P} + b\vec{Q}$ (b) $\vec{S} = (b-1)\vec{P} + b\vec{Q}$

(c) $\vec{S} = (1-b^2)\vec{P} + b\vec{Q}$ (d) $\vec{S} = (1-b)\vec{P} + b^2\vec{Q}$

$\vec{P} + b|\vec{R}|\hat{R} = \vec{S}$

$\vec{P} + b|\vec{R}|\frac{\vec{R}}{|\vec{R}|} = \vec{S}$

$\vec{P} + b\vec{R} = \vec{S}$

$\vec{P} + b(\vec{Q} - \vec{P}) = \vec{S}$

$\vec{P} + b\vec{Q} - b\vec{P} = \vec{S}$

$\vec{S} = \vec{P}(1-b) + b\vec{Q}$

Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\,\hat{i}$ and $\vec{B} = a\,(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j})$, where a is a constant and $\omega = \pi/6$ rad s^{-1}. If $|\vec{A} + \vec{B}| = \sqrt{3}\,|\vec{A} - \vec{B}|$ at time $t = \tau$ for the first time, the value of τ, in seconds, is _____

[JEE(Advance) 2018]

$\vec{A}\cdot\vec{B} = AB\cos\theta$

$a^2\cos(\omega t) = a\times a\times\cos 60$

$\omega t = \pi/3$

$\frac{\pi}{6}t = \pi/3$

$t = 2$ s

(#) $\vec{A} = a\hat{i}$ $\quad |\vec{A}| = a$

$\vec{B} = a\cos(\omega t)\,\hat{i} + a\sin(\omega t)\,\hat{j}$

(#) $|\vec{B}| = \sqrt{a^2(\cos^2\omega t + \sin^2\omega t)} = \sqrt{a} = a$

$|\vec{A}+\vec{B}| = \sqrt{3}\,|\vec{A}-\vec{B}|$

$2a\cos(\theta/2) = \sqrt{3}\,2a\sin\theta/2$

$\tan(\theta/2) = \frac{1}{\sqrt{3}}$

$30^\circ = \frac{\theta}{2}$

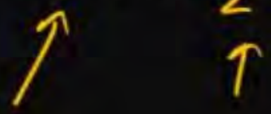
$\theta = 60$

Question JEE

If resultant of two vector is half of bigger vector and perpendicular to smaller vector then find angle between the vector.

Question

Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is: **[JEE Mains 2023]**



1. $\frac{\sqrt{5}A}{4}$
2. $\frac{\sqrt{5}A}{2}$
3. $\frac{5A}{2}$
4. $\frac{\sqrt{5}A^2}{2}$

$$R = \sqrt{A^2 + (A/2)^2 + 2A \cdot \frac{A}{2} \cdot \cos 90^\circ}$$

$$= \sqrt{A^2 + \frac{A^2}{4} + 0}$$

$$= \sqrt{\frac{5A^2}{4}}$$

$$= \frac{A}{2}\sqrt{5}$$

Question

When vector $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ is subtracted from vector $\vec{B}$ it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\vec{B}$ will be:

[JEE Mains 2023]

1. $\sqrt{5}$
2. 3
3. $\sqrt{6}$
4. $\sqrt{33}$

$\vec{B} - \vec{A} = 2\hat{j}$

$\vec{B} = 2\hat{j} + \vec{A}$

$\vec{B} = 2\hat{j} + 2\hat{i} + 3\hat{j} + 2\hat{k}$

$\vec{B} = 2\hat{i} + 5\hat{j} + 2\hat{k}$

$|\vec{B}| = \sqrt{4 + 25 + 4}$

$= \sqrt{33}$

Question

An electric field is given by $(6\hat{\imath} + 5\hat{\jmath} + 3\hat{k})$ N C^{-1}. The electric flux through a surface area $30\hat{\imath}$ m^2 lying in YZ-plane (in SI unit) is:

[JEE Mains 2024]

1. 90
2. 150
3. 180
4. 60

$\vec{E} = 6\hat{i} + 5\hat{j} + 3\hat{k}$

$\vec{A} = 30\hat{i}$

$\phi = \vec{E} \cdot \vec{A}$

$= 180$

Question

If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at an angle θ, then

[JEE Mains 2024]

1. $\vec{A} - \vec{B} = \sqrt{2}R \sin\frac{\theta}{2}$

2. $\vec{A} + \vec{B} = 2R \sin\frac{\theta}{2}$

3. $\vec{A} + \vec{B} = 2R \cos\frac{\theta}{2}$

4. $\vec{A} - \vec{B} = 2R \cos\frac{\theta}{2}$

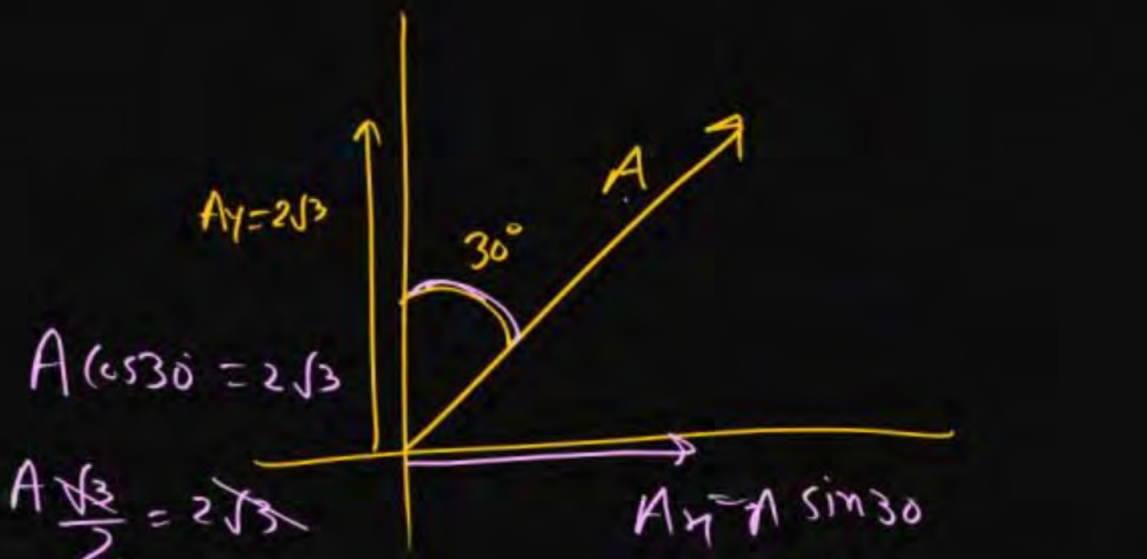
$|\vec{A}| = |\vec{B}| = R$

Question

A vector in x-y plane makes an angle of 30° with y-axis. The magnitude of y-component of vector is $2\sqrt{3}$. The magnitude of x-component of the vector will be:

[JEE Mains 2023]

1. $1/\sqrt{3}$
2. 6
3. 2
4. $\sqrt{3}$



Question

If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other. Then, the value of m will be

[JEE Mains 2023]

1. -1
2. 2
3. 3
4. 1

$4 - 4m + m^2 = 0$

$m^2 - 4m + 4 = 0$

$m^2 - 2m - 2m + 4 = 0$

Question

If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be ____.

Component

[JEE Mains 2021/22]

$\vec{A} \perp_r \vec{B}$

$\vec{A} \cdot \vec{B} = 0$

Question

Three vectors $\overrightarrow{OP}, \overrightarrow{OQ}$ and $\overrightarrow{OR}$ each of magnitude A ($=R$) are acting as shown in figure. The resultant of the three vectors is $A\sqrt{x}$. The value of x is _____. **[JEE Mains 2024]**





$x = 3$

$R_{net} = \sqrt{R^2 + 2R^2 + 2 \times R \times \sqrt{2}R \cos 90^\circ}$

$R\sqrt{x} = \sqrt{3R^2}$

Question

A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}$ N. The value of work done will be _____ J.

[JEE Mains 2023]

final

initial

$W = \vec{F} \cdot \vec{\Delta S}$

$\vec{\Delta S} = \vec{r}_f - \vec{r}_i$

$= (5\hat{i} + 2\hat{j} + 7\hat{k}) \cdot [5\hat{i} - 2\hat{j} + \hat{k} - 2\hat{i} - 3\hat{j} + 4\hat{k}]$

$= (5\hat{i} + 2\hat{j} + 7\hat{k}) \cdot (3\hat{i} - 5\hat{j} + 5\hat{k})$

$= 15 - 10 + 35 =$

$= 40\,J$

Question

A body of mass 4 kg experiences two force $\vec{F}_1 = 5\hat{i} + 8\hat{j} + 7\hat{k}$ and $\vec{F}_2 = 3\hat{i} - 4\hat{j} - 3\hat{k}$.
The acceleration acting on the body is:

[JEE Mains 2024]

1. $-2\hat{i} - \hat{j} - \hat{k}$
2. $4\hat{i} + 2\hat{j} + 2\hat{k}$
3. $2\hat{i} + \hat{j} + \hat{k}$
4. $2\hat{i} + 3\hat{j} + 3\hat{k}$

$$a = \frac{\vec{F_1} + \vec{F_2}}{m}$$

$$a = \frac{8i + 4J + 4K}{4}$$

$$\vec{a} = 2i + J + K$$

Question

copy in Note

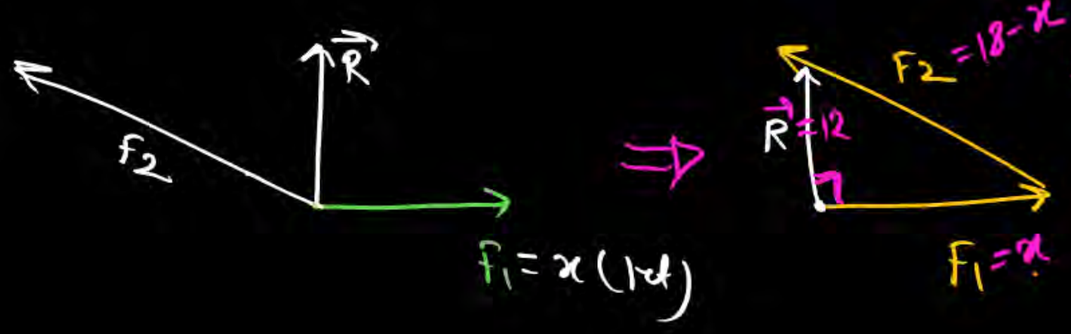
Q. The sum of the magnitudes of two forces acting at point is 18 and the magnitude of their resultant is 12. If the resultant is at 90° with the force of smaller magnitude, what are the magnitudes of forces?

1. 12, 5
2. 14, 4
3. 5, 13
4. 10, 8

$|\vec{F_1}| + |\vec{F_2}| = 18$

$5 + |F_2| = 18 \quad |F_2| = 18 - 5 = 13$

$|\vec{F_1} + \vec{F_2}| = 12$



right angle Triangle.

$x^2 + (12)^2 = (18-x)^2$

↓ find x

$x^2 + 144 = 324 + x^2 - 36x$

$x = 5$

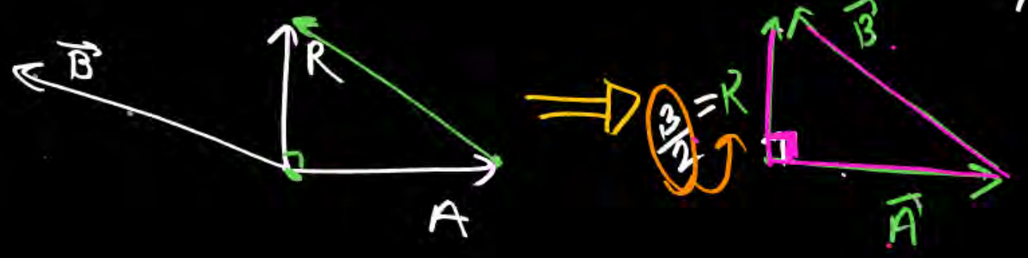
18 × 18 = 324, 6

(Q) If Resultant of two vector is ⊥r to Smaller vector and half in magnitude of bigger vector then find Small vector.

Sol^n

$$\vec{R} = \vec{A} + \vec{B}$$
(Small, large)

$$|\vec{R}| = \frac{|B|}{2}$$



Pythag

$$A^2 + \left(\frac{B}{2}\right)^2 = B^2$$

$$A^2 + \frac{B^2}{4} = B^2$$

$$A^2 = B^2 - \frac{B^2}{4}$$

$$A^2 = \frac{3B^2}{4}$$

$$A = \frac{\sqrt{3}B}{2} = \sqrt{3}R$$

Question

MR Scam ✓ Copy

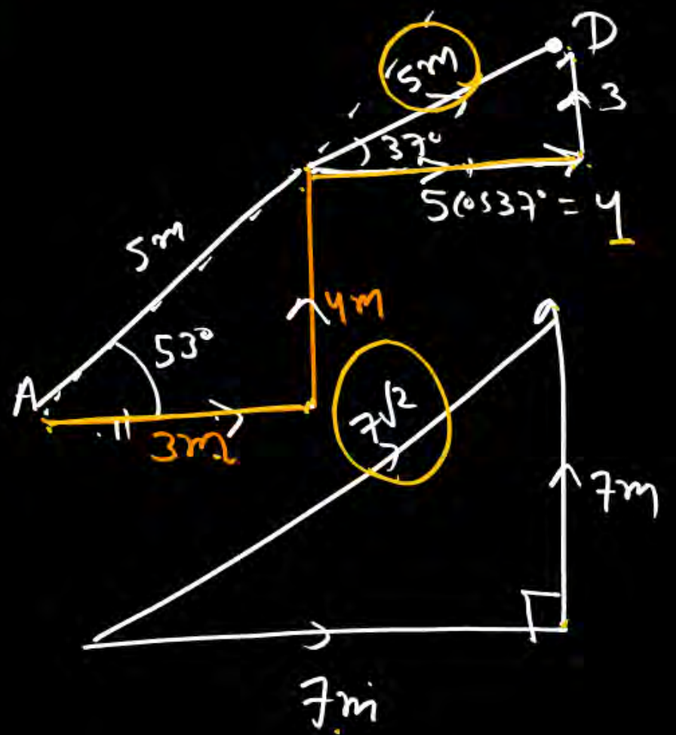
A particle moves along a path ABCD as shown in the figure. Then the magnitude of net displacement of the particle from position A to D is:

1. 10 m (7√2)
2. $5\sqrt{2}$ m
3. 9 m
4. $7\sqrt{2}$ m

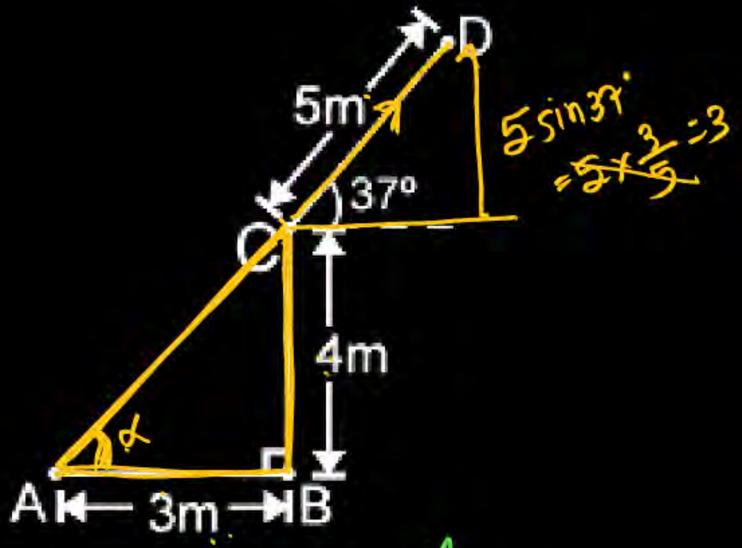
5% Notukka

add one pen in dbt.

Soln



Soln ↓



$\tan\alpha = \frac{4}{3}$

$\alpha = 53°$ ✓

Question

Which of the following combination of three force can give zero resultant?

1. [2, 4, 7]
2. [3, 1, 5]
3. [2, 8, 11]
4. [3, 4, 2]

Question

Conceptual

For two vector $\vec{A}$ and $\vec{B}$ vector addition $\vec{R} = \vec{A} + \vec{B}$ and vector subtraction $\vec{D} = \vec{A} - \vec{B}$ then angle between $\vec{A}$ and $\vec{B}$.

Condition	Angle
(i) $\lvert\vec{A}+\vec{B}\rvert = \lvert\vec{R}\rvert = \lvert\vec{A}\rvert = \lvert\vec{B}\rvert$	$\rightarrow 120^\circ$
(ii) $\lvert\vec{A}\rvert = \lvert\vec{B}\rvert = \lvert\vec{A}-\vec{B}\rvert$	$\rightarrow 60^\circ$ ✓
(iii) $A^2 + B^2 = \lvert\vec{A}-\vec{B}\rvert^2 = D^2$	$\rightarrow 90^\circ$
(iv) $A^2 + B^2 = R^2$	$\rightarrow 90^\circ$

$\theta = 90^\circ$

$R = \sqrt{A^2+B^2+2AB\cos\theta}$

$R^2 = A^2+B^2+2AB\cos\theta$

$A^2+B^2 = A^2+B^2+2AB\cos\theta$

$0 = 2AB\cos\theta$ $\cdot\cos\theta = 0^\circ$

$\theta = 90^\circ$

$|\vec{A}| = A \quad |\vec{B}| = B$

$|\vec{A}+\vec{B}| = |\vec{R}| = R$

$|\vec{A}-\vec{B}| = |\vec{D}|$

Question

Condition		Angle
(v) $\lvert\vec{A}+\vec{B}\rvert = \lvert\vec{A}-\vec{B}\rvert$	→	90° ✓
(vi) $\lvert\vec{A}+\vec{B}\rvert = \lvert\vec{A}\rvert + \lvert\vec{B}\rvert$	→	0°
(vii) $\lvert\vec{A}-\vec{B}\rvert = \lvert\vec{A}\rvert + \lvert\vec{B}\rvert$	→	180° ✗
(viii) $\lvert\vec{A}+\vec{B}\rvert = \lvert\vec{A}\rvert - \lvert\vec{B}\rvert$	→	(80
(ix) $\lvert\vec{A}-\vec{B}\rvert = \lvert\vec{A}\rvert - \lvert\vec{B}\rvert$	→	0°

$R = \sqrt{A^2+B^2+2AB\cos\theta}$

$D = \sqrt{A^2+B^2-2AB\cos\theta}$

Compare

$R = A+B$ ✓

$D = \sqrt{A^2+B^2-2AB\cos\theta}$ ✓

$\theta = 180°$

$= \sqrt{A^2+B^2+2AB\times 1}$

$= \sqrt{(A+B)^2} = A+B$

Question

Find magnitude of Vector:

$\vec{A} = 2\hat{i} + 3\hat{j}$ → $|\vec{A}| = \sqrt{4+9} = \sqrt{13}$

$\vec{B} = 3\hat{i} + 4\hat{j}$ → B = S.

$\vec{C} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ → $|C| = \sqrt{3^2+4^2+(5)^2} = \sqrt{9+16+25} = \sqrt{50} = \sqrt{25\times 2} = 5\sqrt{2}$

$\vec{D} = \hat{i} - \hat{j} + \hat{k}$ → $|\vec{D}| = \sqrt{1^2+(-1)^2+(1)^2} = \sqrt{3}$

$\vec{E} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ →

$\vec{F} = 10\hat{i} - 10\hat{j} - 10\hat{k}$ → $|\vec{F}| = \sqrt{(10)^2+(-10)^2+(-10)^2} = \sqrt{100+100+100} = \sqrt{3\times 100} = \sqrt{3}\times\sqrt{3\times 100}$

$= 10\sqrt{3}$

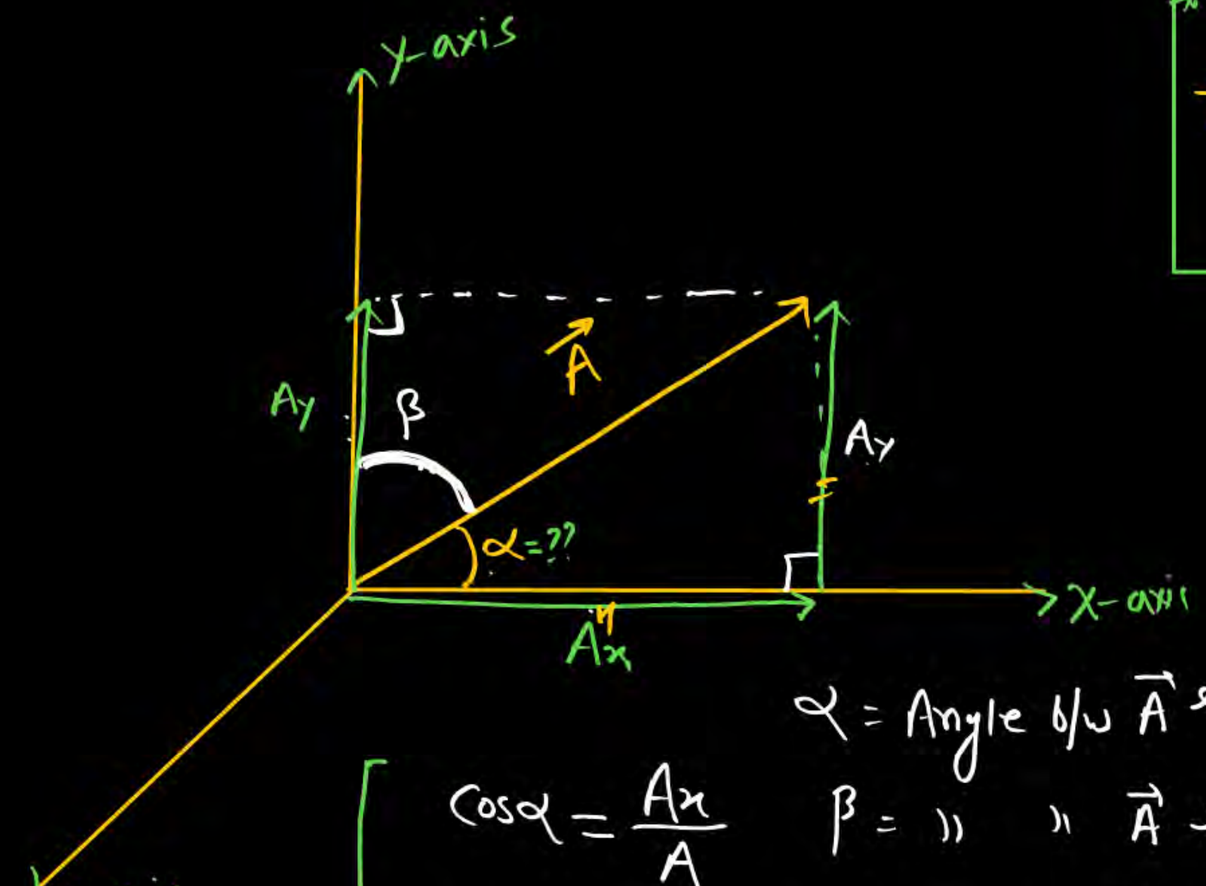
$$\vec{A} = \sin\theta\, i - \cos\theta\, \hat{j}$$

$$|\vec{A}| = \sqrt{\sin^2\theta + \cos^2\theta} = \sqrt{1} = 1$$

(a) If $\vec{c} = 2i + 3\hat{j} - 4\hat{k}$ find unit vector of $\vec{c}$.

$$\hat{c} = \frac{\vec{c}}{c} = \frac{2i + 3\hat{j} - 4\hat{k}}{\sqrt{4+9+16}} = \frac{2i + 3\hat{j} - 4\hat{k}}{\sqrt{29}}$$

Find Angle b/w vector and x-axis, y-axis & z-axis.



α = Angle b/w $\vec{A}$ & x-axis

β = " " $\vec{A}$ & Y-axis.

γ = " " $\vec{A}$ & Z-axis.

$$\begin{cases} \cos\alpha = \dfrac{A_x}{A} \\ \cos\beta = \dfrac{A_y}{A} \\ \cos\gamma = \dfrac{A_z}{A} \end{cases}$$

$$\tan\alpha = \frac{A_y}{A_x}$$

↳ Angle b/w $\vec{A}$ with x-axis

(Q) find $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = ??$

$$\frac{A_x^2}{A^2} + \frac{A_y^2}{A^2} + \frac{A_z^2}{A^2} =$$

$$\frac{A_x^2 + A_y^2 + A_z^2}{A^2} = \frac{A^2}{A^2} = 1$$

find

$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = ??$

$1 - \cos^2\alpha + 1 - \cos^2\beta + 1 - \cos^2\gamma = 3 - [1]$

$\rightarrow 2$

Question

(θ) copy

Vector $\vec{A} = 5\hat{i} + 5\sqrt{3}\hat{j}$ then find angle of vector $\vec{A}$ from x-axis.

$\vec{A} = 5\hat{i} + 5\sqrt{3}\hat{j}$

m-2

$\cos\alpha = \frac{A_x}{A} = \frac{5}{\sqrt{5^2+(5\sqrt{3})^2}} = \frac{5}{\sqrt{25+75}} = \frac{5}{\sqrt{100}}$

$\cos\alpha = \frac{1}{2}$

$\alpha = 60°$

m-2

$\tan\alpha = \frac{A_y}{A_x}$

$\tan\alpha = \frac{5\sqrt{3}}{5} = \sqrt{3}$

$\alpha = 60°$

$\cos\gamma = \frac{A_z}{A} = \frac{0}{10} = 0$

$\cos\gamma = 0$

$\gamma = 90°$

Question

Copy

The angles which a vector $\hat{i} + \hat{j} + \sqrt{2}\,\hat{k}$ makes with X, Y and Z axes respectively are

1. $60°, 60°, 60°$
2. $45°, 45°, 45°$
3. $60°, 60°, 45°$
4. $45°, 45°, 60°$

$|A| = \sqrt{1^2+1^2+(\sqrt{2})^2}$

$= \sqrt{4} = 2$

$\cos\alpha = \frac{1}{2} \quad \alpha = 60$

$\beta = 60$

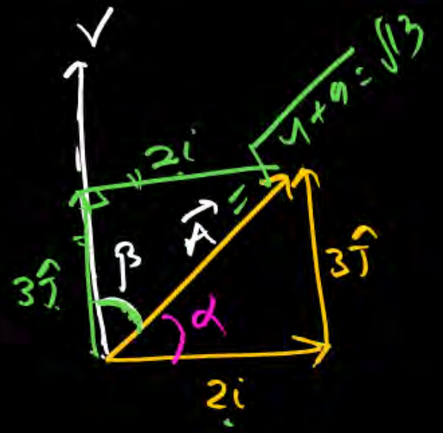
$\cos\gamma = \frac{\sqrt{2}}{2}$

$\gamma = 45$

Question

Given $\vec{A} = 2\hat{i} + 3\hat{j}$, the angle between $\vec{A}$ and Y-axis is

1. $\sin^{-1}\frac{2}{3}$
2. $\cos^{-1}\frac{2}{3}$
3. $\tan^{-1}\frac{2}{3}$
4. $\tan^{-1}\frac{3}{2}$



$\tan\alpha = \frac{3}{2}$

$\alpha = \tan^{-1}(3/2)$

$\tan\beta = \frac{2}{3}$

$\beta = \tan^{-1}\left(\frac{2}{3}\right)$

$COR = \frac{2}{\sqrt{13}}$

Question

H/W →

If $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} + 3\hat{k}$ find:

(i) $\vec{A} + \vec{B}$

(ii) $\vec{B} - \vec{A}$

(iii) $\vec{B} - \frac{\vec{A}}{2}$ $= i + 2\hat{j} + 3\hat{k} - \left(\frac{2i + 3\hat{j} + 4k}{2}\right)$

(iv) $2\vec{A} + \vec{B}$ $= (i - i) + \left(2\hat{j} - \frac{3}{2}\hat{j}\right) + (3\hat{k} - 2\hat{k})$

(v) $\vec{A} - 2\vec{B}$

(vi) $|\vec{A}| + |\vec{B}|$

(vii) $\frac{|\vec{A}|}{|\vec{B}|}$

(viii) $\frac{\vec{A}}{\vec{B}}$ = Not Possible ✓

Question

If the sum of two unit vectors is a unit vector, then the magnitude of their difference is:

1. $\sqrt{3}$
2. $\sqrt{2}$
3. $\sqrt{5}$
4. $1/\sqrt{2}$

Checking Unit Vector

$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$



By finding magnitude

$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} = 1$

then Unit Vector

$\neq 1$ Not a Unit Vector

$\vec{A} \cdot \vec{A} = 1$ $= 1 \times 1 \cos 0^\circ = 1$

then $\vec{A}$ is Unit Vector.

$\vec{A} = 2\hat{i}$

$\vec{B} = 2\hat{j}$

$\frac{\vec{A}}{\vec{B}} = \frac{2\hat{i}}{2\hat{j}} = \frac{\hat{i}}{\hat{j}}$

Question

If $\vec{A} = 0.6\hat{\imath} + \beta\,\hat{\jmath}$ is a unit vector then, find value of 'β'?

Not Copy

1. 0.4
2. 0.8
3. 0.6
4. 0.7

$\vec{A} = 0.6\hat{\imath} + \beta\hat{\imath}$

$\vec{A} \cdot \vec{A} = 1$

$0.6 \times 0.6 + \beta^2 = 1$

$\beta^2 = 1 - 0.36$

$\beta^2 = 0.64$

$\beta = \sqrt{0.64} = \sqrt{\frac{64}{100}}$

$= \frac{8}{10} = 0.8$

Question

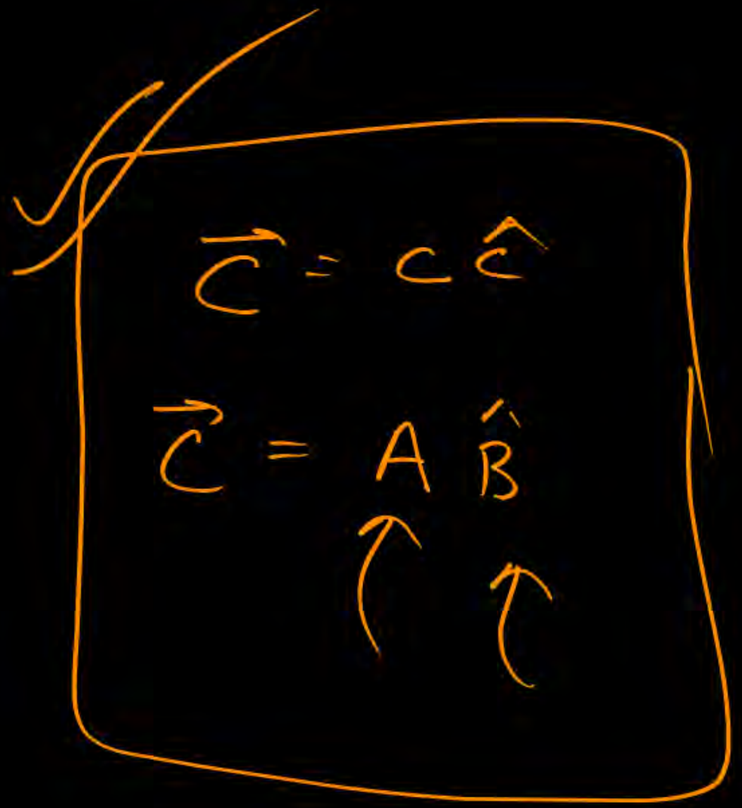
Find value of β if $\vec{A}$ is *a* unit vector $\vec{A} = 0.4\hat{\imath} + 0.3\hat{\jmath} + \beta\hat{k}$.



Question

H/W

If $\vec{A} = 2\hat{\imath} + \hat{\jmath} + 2\hat{k}$ and $\vec{B} = \hat{\imath} + \hat{\jmath} + 2\hat{k}$ then find direction $\vec{C}$ magnitude same as $\vec{A}$ and direction along $\vec{B}$.



Question

Three forces given by vectors $2\hat{\imath} + 2\hat{\jmath}; -2\hat{\imath} - 2\hat{\jmath}$ and $-4\hat{\imath}$ are acting together on a point object at rest. The object moves along the direction

1. x-axis
2. y-axis
3. z-axis
4. Object does not move

Question

Two vector $\vec{A} = 6\hat{\imath}$ and $\vec{B} = 8\hat{\jmath}$ then find $\vec{A} + \vec{B}$, direction of $\vec{A} + \vec{B}$ from $\vec{A}$.

Thank You